

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715
SARANYA.R.A(192572052)

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

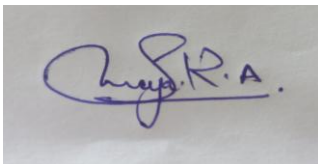
ARTICLE REVIEW

Code of Conduct:

I Saranya.R.A(192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

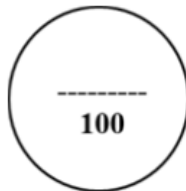
SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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CO4 – ASSESSMENT TOOL 3

Article Review

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17
QUESTIONS: 4 **Duration:** 1 Hour **Total Marks:** 40

Criteria	Weightage	Task 1 (10)	Task 2 (10)	Task 3 (10)	Task 4 (10)	Total Marks (40)
Understanding of Article	20% (2)	/2	/2	/2	/2	/8
Summary	20% (2)	/2	/2	/2	/2	/8
Critical Analysis	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Use of Evidence	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/6
Organization	20% (2)	/2	/2	/2	/2	/8
Total per Question	10 Marks	/10	/10	/10	/10	/40



ANSWER

Article Title: *Nutritional intake of micronutrient and macronutrient and type 2 diabetes: machine learning schemes*

Authors: Mohammad Rashidmayvan, Amin Mansoori, Elahe Derakhshan-Nezhad, Davoud Tanbakuchi, Fatemeh Sangin, Maryam Mohammadi-Bajgiran, Malihehsadat Abedsaeidi, Sara Ghazizadeh, MohammadReza Mohammad Taghizadeh Sarabi, Ali Rezaee, Gordon Ferns, Habibollah Esmaily, and Majid Ghayour-Mobarhan.

Journal: *Journal of Health, Population and Nutrition*

Year: 2025, Volume 44, Article 31

TASK 1: ARTICLE SUMMARY

(a) Domain and ML Problem

Domain: Healthcare, nutrition and diabetes prediction.

Application: Early identification of risk factors associated with Type 2 Diabetes (T2D).

Machine Learning Problem: The authors investigate whether micronutrient and macronutrient intake can be used to identify factors associated with the development of Type 2 Diabetes.

The study uses two CO 4 techniques:

- **Decision Tree:** Uses the Gini impurity index to identify important nutritional variables and generate interpretable decision rules.
- **Logistic Regression:** Estimates the relationship between nutritional variables and the probability of T2D.

The research used data from the **Mashhad Stroke and Heart Atherosclerotic Disorders (MASHAD) cohort study** in Iran. The original study population contained 9,704 individuals aged 35–65 years, with long-term follow-up for determining T2D incidence. After data cleaning and balancing, 9,000 participants remained, with 4,500 developing T2D and 4,500 remaining non-diabetic.

(b) Key Objective and Research Gap — 150–200 Words

The main objective of the article is to investigate whether dietary micronutrient and macronutrient intake can help identify the risk of developing Type 2 Diabetes. The authors focus on nutritional factors because diabetes is strongly associated with lifestyle and dietary behaviour, but previous studies have not consistently established the contribution of individual nutrients to diabetes incidence. The authors identify a research gap in the limited number of large Asian studies examining these relationships using long-term follow-up data. They particularly note that many previous investigations used relatively small populations or short follow-up periods. To address this gap, the researchers analysed the large MASHAD cohort from Iran, which followed participants for approximately ten years. They applied Logistic

Regression and Decision Tree models to investigate associations between nutritional variables and T2D. The Decision Tree used the Gini impurity index to identify important variables and generate interpretable rules, while Logistic Regression quantified statistical associations through regression coefficients and odds ratios. The study found important relationships involving potassium, tryptophan, protein, carbohydrates, cholesterol and other nutrients.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

1. Decision Tree

The authors used a **Decision Tree classification approach** based on the **Gini impurity index**.

The Gini impurity is:

$$Gini(D) = 1 - \sum_{i=1}^m P_i^2$$

The algorithm selects variables that provide useful separation between T2D and non-T2D groups. The resulting tree was subsequently pruned.

For micronutrient variables, the Decision Tree produced **5 layers and 14 splits**. The most important variables were:

1. Potassium
2. Tryptophan
3. Iodine
4. Sodium
5. Riboflavin

For macronutrients, the important variables were:

1. Protein
2. Carbohydrate
3. Cholesterol
4. Fiber

2. Logistic Regression

Logistic Regression was used as a statistical learning model for the binary outcome:

T2D = Yes/No

The model examined nutritional variables including:

- Selenium
- Magnesium
- Potassium
- Vitamin D

- Zinc
- Calcium
- Iron
- Iodine
- Tryptophan
- Vitamin B12
- Protein
- Carbohydrates
- Fiber
- Cholesterol
- Different types of fatty acids

The authors used regression coefficients, p-values and odds ratios to identify statistically significant nutritional predictors.

Dataset

Dataset characteristic Details

Dataset source	MASHAD Cohort Study, Iran
Original participants	9,704
Final cleaned dataset	9,000
Age	35–65 years
Follow-up	Approximately 10 years
T2D cases	4,500
Non-T2D cases	4,500
Main features	Micro- and macronutrient intake
Outcome	Incidence of Type 2 Diabetes

The researchers collected blood samples after fasting and used questionnaire and clinical information as part of the cohort data.

(b) Mapping to CO 4

CO 4 Topic	Application in Article
Inductive Learning	General patterns were learned from existing patient/nutritional records to identify T2D risk factors.
Decision Trees	Gini impurity was used to construct an interpretable tree and identify important variables.
Statistical Learning	Logistic Regression was used to estimate relationships between

nutritional variables and T2D.

Reinforcement Learning

Not used in this article.

Methodological Strength

A major strength is the **large sample size and approximately 10-year follow-up**, which provide substantially more information than a small cross-sectional dataset. The use of both Decision Tree and Logistic Regression also provides complementary perspectives: interpretable rules from the tree and statistical associations from regression.

Methodological Limitation

The authors acknowledge that the analysis did not adjust for potentially important confounding variables such as **smoking, medication use, BMI, family history of diabetes and sex**. This can affect the interpretation of nutritional variables and may introduce confounding bias.

TASK 3: RESULTS AND CRITICAL EVALUATION

(a) Key Results

An important point is that this article is primarily an **association/risk-factor analysis**, rather than a conventional benchmark study reporting classification accuracy, F1-score and AUC for competing predictive models. Therefore, it would be incorrect to invent accuracy or AUC values.

Decision Tree Results

Feature group	Most important variables	Reported finding
Micronutrients	Potassium, Tryptophan, Iodine, Sodium, Riboflavin	These variables formed the most important Decision Tree splits.
Macronutrients	Protein, Carbohydrate, Cholesterol, Fiber	These were the major Decision Tree predictors.

The micronutrient Decision Tree had **5 layers and 14 splits**. In one reported subgroup, high potassium combined with high iodine corresponded to an approximately **80% diabetic proportion**. Another subgroup with low potassium, low tryptophan, low sodium, low magnesium and low zinc had approximately **97% non-diabetic participants**.

For macronutrients, the Decision Tree identified protein, carbohydrate, cholesterol and fiber as the most important factors. A subgroup with high protein and carbohydrate intake had a reported diabetes-development proportion of approximately **94%**.

Logistic Regression Results

Logistic Regression identified significant relationships involving several micronutrients, including:

- Potassium

- Tryptophan
- Sodium
- Chloride
- Phosphate
- Carotene
- Iodine
- Vitamin B12
- Iron
- Calcium
- Riboflavin
- Vitamin D
- Zinc
- Magnesium

Among macronutrients, carbohydrate, water, protein, cholesterol, saturated fat, monounsaturated fat and fiber showed significant associations.

(b) Critical Evaluation

Are the Results Realistic?

The findings are plausible because nutritional intake can be associated with metabolic health and diabetes risk. However, the reported associations should **not be interpreted as proof that changing one nutrient alone will prevent diabetes.**

The study itself reports that relationships can differ across populations and previous studies have sometimes produced conflicting findings, particularly for nutrients such as iodine, vitamin D, protein and zinc.

Potential Biases

1. Population bias:

The study is based on the MASHAD cohort in Iran, so the results may not generalise directly to all populations.

2. Confounding:

Important variables such as BMI, smoking, medication use, family history and sex were not fully adjusted for in the reported analysis.

3. Dietary measurement bias:

Nutritional intake can be difficult to measure accurately because it may depend on participant recall and reporting.

4. Association versus causation:

A statistical association does not establish that a particular nutrient directly causes or prevents diabetes.

Additional Experiments

The study could be strengthened by:

- External validation using another country's cohort.
- Repeated dietary measurements over time.
- Including BMI, physical activity, medication use and family history.
- Comparing Decision Tree, Random Forest, XGBoost and Logistic Regression.
- Reporting AUC-ROC, precision, recall and F1-score if prediction is a primary goal.
- Using cross-validation to estimate model stability.
- Applying explainability methods such as SHAP for more detailed interpretation.

(c) Real-World Applicability

The approach could potentially support:

- Diabetes risk screening.
- Nutrition-risk assessment.
- Electronic Health Record decision support.
- Population-level diabetes prevention programmes.
- Personalised nutrition research.
- Public-health risk stratification.

Industry Challenges

Data availability:

Healthcare organisations require sufficiently large and high-quality longitudinal datasets.

Data integration:

Dietary information, laboratory measurements and clinical records may exist in different systems.

Generalisation:

A model developed from an Iranian cohort may not perform identically in Indian, European or other populations.

Ethical issues:

Healthcare AI must avoid unfair predictions and should not replace qualified clinical judgement.

Interpretability:

Clinical users need understandable reasons for predictions rather than unexplained model outputs.

Validation:

Independent external validation is necessary before clinical deployment.

TASK 4: REFLECTION AND LEARNING OUTCOME

(a) Three Key Insights

Insight 1 — Decision Trees provide interpretable rules

I learned that a Decision Tree does more than classify data; it can generate understandable decision rules by repeatedly splitting the dataset using important features.

CO 4 connection: Decision Trees and Gini Impurity.

The article demonstrated that potassium, tryptophan, iodine and other variables could be arranged into interpretable decision paths.

Insight 2 — Logistic Regression provides statistical interpretation

The article helped me understand that Logistic Regression can provide more than a class prediction. Its coefficients, p-values and odds ratios help explain the relationship between input variables and a binary outcome.

CO 4 connection: Statistical Learning — Logistic Regression.

For example, the researchers used odds ratios to interpret the relationship between nutritional variables and T2D.

Insight 3 — Large longitudinal datasets improve healthcare analysis

The approximately 10-year follow-up demonstrates how longitudinal data can provide information about disease incidence rather than relying only on a single measurement.

CO 4 connection: Inductive Learning and supervised ML.

The study shows how historical observations can be used to learn patterns associated with future T2D incidence.

(b) Proposed Novel Extension

Proposed Experiment: External Validation Using an Indian Diabetes Dataset

I would extend the research by testing the same Decision Tree and Logistic Regression methodology on a large **Indian longitudinal diabetes dataset** containing age, BMI, glucose, blood pressure, family history, physical activity and dietary information.

The experiment would use:

1. Data cleaning and missing-value treatment.
2. Stratified train-test splitting.
3. Feature normalisation where appropriate.
4. Class-imbalance handling.
5. Decision Tree with cost-complexity pruning.
6. Logistic Regression with regularisation.
7. 5-fold cross-validation.
8. Accuracy, precision, recall, F1-score and AUC-ROC.
9. SHAP-based feature interpretation.
10. External validation on a separate population.

Expected Impact

This extension would determine whether the nutritional predictors identified in the original Iranian cohort generalise to another population. It would also transform the study from primarily an association analysis into a more comprehensive **predictive modelling study**. If the model maintains good recall and AUC on an independent population, it would provide stronger evidence for its potential use in diabetes risk-screening systems.

Overall Review and Conclusion

The selected article is highly relevant to **CO 4** because it applies both **Decision Tree learning and Logistic Regression** to a real healthcare problem. Its major strengths are the large cohort, long follow-up period and interpretable machine-learning methods. The Decision Tree identified important nutritional variables, while Logistic Regression provided statistical measures such as coefficients and odds ratios.

However, the study should be interpreted carefully because it does not primarily report conventional predictive metrics such as accuracy, F1-score and AUC-ROC, and the authors acknowledge important unadjusted confounding factors. Therefore, additional external validation, broader feature inclusion and standard predictive evaluation would strengthen the research.

Overall, the article demonstrates how **Inductive Learning, Decision Trees and Statistical Learning** can be applied to healthcare data to discover meaningful patterns and support diabetes-risk research.